

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

TEAM ID	SWTID2026-3387
PROJECT TITLE	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
MAX MARKS	5 MARKS

4.2 – Proposed Solution

Proposed Solution Using Tableau

The proposed solution is to develop an interactive business intelligence system using **Tableau** that analyzes housing market data and presents meaningful insights through dynamic dashboards and visualizations. The system helps users understand how different property features influence house sale prices, identify market trends, and make data-driven decisions. Instead of relying on manual analysis of large datasets, Tableau provides an intuitive platform for exploring housing data using charts, filters, and dashboards.

The solution begins by collecting housing market data from sources such as Excel files, CSV files, or SQL databases. The dataset includes important attributes such as sale price, number of bedrooms, bathrooms, house area, basement area, year built, property condition, grade, and location. Tableau connects directly to these data sources and imports the data for analysis.

Once the data is imported, it is cleaned and prepared by removing duplicate records, handling missing values, correcting data types, and creating calculated fields where necessary. This preprocessing ensures that the dataset is accurate, consistent, and suitable for visualization.

Tableau is then used to create a variety of interactive visualizations that help users analyze housing market trends. Bar charts compare average sale prices across different categories, line charts display price trends over time, scatter plots reveal relationships between house size and sale price, pie charts show category-wise distributions, histograms analyze the frequency of property prices, and heat maps identify high-value housing locations. These visualizations make complex housing data easy to understand.

The individual worksheets are combined into an interactive dashboard that provides a comprehensive view of the housing market. Users can apply filters based on location, number of bedrooms, house condition, year built, or other features to explore specific

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segments of the market. Interactive tooltips and highlighting actions further improve the user experience by providing additional information without cluttering the dashboard.

A Tableau Story is created to present the analysis in a logical sequence. The storyboard begins with an overview of the dataset, followed by detailed analysis of sale prices, property characteristics, market trends, and key findings. This storytelling approach helps stakeholders understand the analytical process and interpret the results more effectively.

To make the solution widely accessible, the Tableau dashboard is published and integrated into a Flask-based web application. This allows users to access the dashboard through a web browser without requiring Tableau Desktop. The web interface is designed to be responsive and user-friendly, ensuring compatibility across different devices.

Performance testing is conducted to verify that the dashboard loads quickly, responds efficiently to user interactions, and produces accurate analytical results. Finally, complete project documentation and demonstrations are prepared to explain the methodology, implementation process, dashboard features, results, conclusions, and future enhancements.

Key Features of the Proposed Solution

- Collects and analyzes housing market data from multiple sources.
- Cleans and prepares data for accurate analysis.
- Creates interactive Tableau visualizations.
- Identifies trends in housing prices and property features.
- Supports filtering, sorting, and drill-down analysis.
- Develops a comprehensive and user-friendly dashboard.
- Uses Tableau Story to present analytical findings.
- Integrates the dashboard into a Flask web application.
- Performs performance testing for reliability and efficiency.
- Provides meaningful insights to support informed real estate decision-making.

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